Gelatinized Starch Investigation

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1 Introduction

This experiment explores the effects of heat and water on starches. A starch is a polysaccharide—a long chain of sugar molecules [1]. It is made up of both amylose and amylopectin, and each type of starch contains a different percentage of each [2]. As a starch is heated to a particular temperature (between 62°C and 75°C for most types of starch), it begins to undergo gelatinization [5]. According to J.M.V. Blanshard of the University of Nottingham, "Gelatinization refers loosely both to the loss of order (...) and also to the swelling of the granule" [2]. The loss of order mentioned here is the breaking down of the starch chain that allows for water absorption and disruption of the crystalline structure of the starch [5]. Gelatinization is important because it changes the taste and texture of a starch granule and makes it easier to digest [4]. We will study the process of gelatinization, how it happens, and under what conditions it occurs. Below are some pictures taken with a digital microscope of potato starch before and after cooking. Figure 1 shows a raw slice of potato with an evident crystalline structure. Figure 2 shows what the starch granules look like after they have absorbed some water and expanded. (Note: Taking some time to use the microscope offers a very helpful visual component for the experiment.)

Studies previously done on the gelatinization of starches have shown mixed results. Experiments by Chi Kai show a linear relationship between time and water uptake by rice as

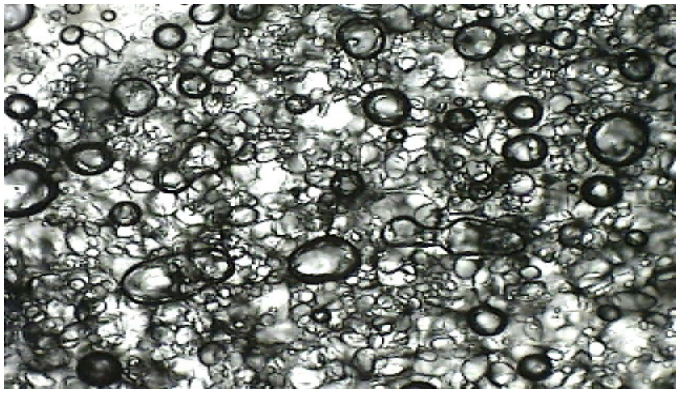

Figure 1: Raw potato slice viewed at 100 times magnification

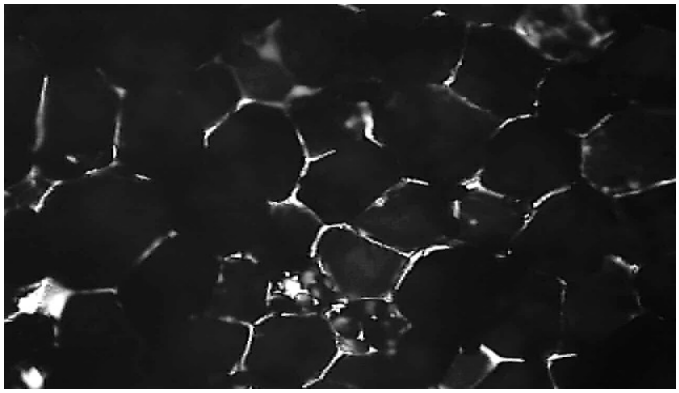

Figure 2: Potato slice cooked in a microwave for 2 minutes in water at 100 times magnification

it is soaked and heated [3]. Other results include those of Landman and McGuiness in their modeling of the cooking of a cereal grain, which is clearly nonlinear [4]. Our preliminary results also indicate a nonlinear relationship. Other experiments have been done with starches to measure heat transfer over cooking time. The results of these agree with our heat transfer model and show a nonlinear relationship between time and percentage gelatinized [6].

We use both rice and potatoes to explore starch gelatinization and water absorption. We are interested in the change of void fraction of rice after it has been heated and absorbed water to its maximum capacity, as this may affects our model of water absorption. Additionally, we will work with potatoes and heat transfer, as well as percentage cooked over time. Our goal is to find a predictive model of the effects of gelatinization. Ready, Set, Cook!

2 Safety

Caution: Water in tank may be very hot. Also, be very careful when using the microtome, as the blades are very sharp.

3 Microscopy: A Visual Component

Learning to use a digital microscope will be very helpful in this investigation. Below are a few basic tips for using a digital microscope:

1. Make sure you have plenty of slides and slide covers.
2. Select a slide and lay your specimen on it.
3. Place cover on a wet sample. You do not need a cover on a dry sample.
4. Place slide onto microscope plate and secure with lens holders on either side of the plate.
5. Turn on microscope. Use the knob on the right side of the plate to bring your sample into focus.

6. To capture your pictures on the computer, bring up the Motic Images Software and make sure the microscope is plugged into the computer. Click on File, Capture, and then save your picture in an appropriate folder.

Additional tips for preparing slides include learning to use the microtome (used to make very thin slices of a material):

1. Remove the blades from the underneath portion of the base.
2. Choose a blade and remove the washer from the blade container as well.
3. Attach blade to left side of slicer with the blade first, then the washer, then the screw found on the side.
4. Cut your potato until you have a long slender strip. It should be about two inches by one half inch by one half inch.
5. Load your potato strip into the moving portion of the slicer by turning the knob on top until the opening is large enough hold the potato strip. Clamp down gently on the potato, making sure the potato comes about one half inch out of the left side.
6. Reattach the moving component and make a large slice. Now you are ready to begin making very thin slices by turning the dial on the right side in very small increments until your slices are of desired thickness.
7. Always clean the slicer blade very soon after use.

4 Experiment 1: Volume Increase of Rice

This experiment will measure the volume increase of rice over a period of three hours due to water absorption and/or heat. We will have two graduated cylinders of rice; one of them will be heated during the experiment. It will require use of the following equipment:

- Large tank with heat capacity of at least 70°C
- Heating utility that attaches to side of tank
- 2 large graduated cylinders
- Plenty of Uncle Ben's Converted Parboiled Enriched Long Grain Rice
- Stopwatch.

4.1 Procedure

Begin by assembling and preparing the water bath:

1. Fasten the water heating utility to the side of the large tank.
2. Fill the tank with water until the water reaches 2 inches from the top.
3. Flip switch in back of heating utility to turn on the unit's power source.
4. Press power button on the front of the unit to turn the unit on.
5. If necessary, use a flathead screwdriver to adjust the "safety set" temperature on the small dial in the lower left corner of the face of the unit.
6. Press P2 and P3 at the same time and wait until the "high temperature" value appears on the screen. Set this value just a few degrees higher than the capacity of the tank using the dial to the right of the screen. Now press P2 and P3 at the same time, twice, to return the screen to the normal temperature view setting (This control turns the heater off should the water temperature rise higher than the tank is capable of holding or if the water level drops below the heating coils.)
7. Press the dial to the right of the temperature viewer in once, and use the dial to adjust your target temperature, which should be 70°C, the maximum heat capacity of the acrylic tank. Cover the tank with aluminum foil. This will help the tank heat up faster and retain its heat throughout your experiments and prevent evaporation.
8. Note: The tank may take up to an hour to heat to the desired temperature, so be sure to turn it on long before you need to use it.

Next, prepare the rice for "cooking":

1. Begin filling both cylinders with rice until the height–which we will now use to refer to volume–of the rice is 50mL.
2. Fill a large beaker with water to add to the rice for cooking. (Do not add the water to the rice until the bath is heated so the rice does not begin absorbing water before the experiment begins.)

When the water bath is heated to 70°C,

- Three things must happen all at the same time (or as close to it as possible):
    1. Pour 250mL of room temperature water into the cylinder that will not be heated during the experiment and 250mL of water directly from the heated water bath into the cylinder that will be heated.
    2. Place the cylinder with hot water into the water bath, making sure it is not touching the sides of the container or the heating utility.
    3. Start the stopwatch.
- Record time and temperature of each cylinder at one-minute intervals for the first hour and ten-minute intervals for the next two hours.

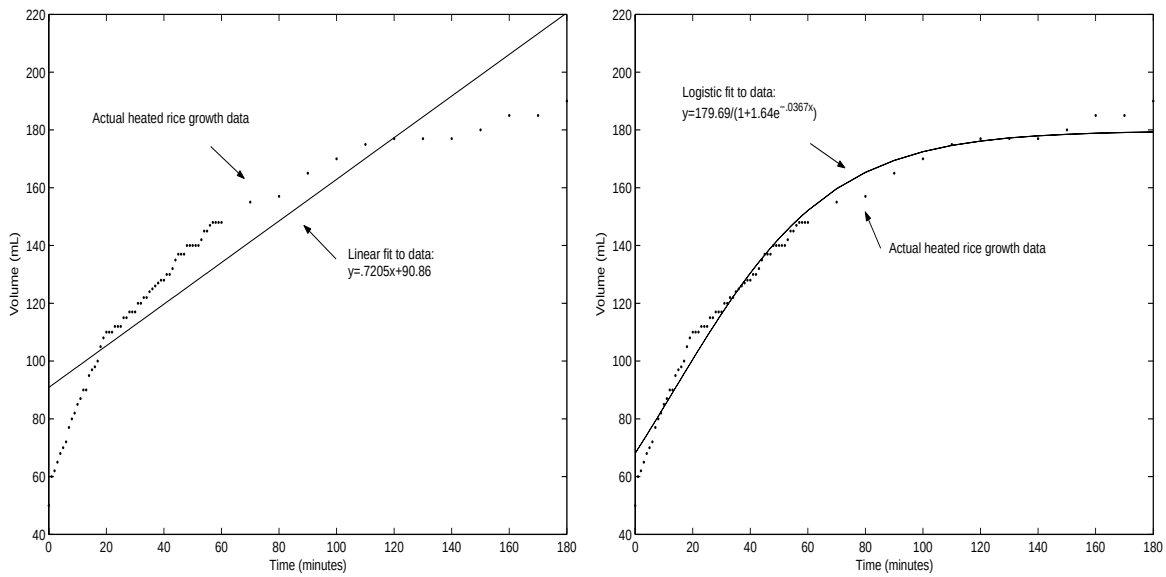

Figure 3: At the left is a linear least squares fit to my data; on the right is the logistic fit. Clearly, the logistic fit is much more accurate!

4.2 Results

After measuring growth each minute for three hours, we used Matlab to graph our data, time versus height. We fit the data to a linear model and a logistic model to compare how well each type of relationship resembled our graphs. Figure 3 shows graphs of the actual data in comparison to the fitted lines. It is clear that the logistic model fits the results much better. We can conclude that water uptake by rice over time is not a linear relationship and better conforms to a logistic curve.

4.3 The Mathematics

Our simple mathematical model of growth of rice over time is a differential equation that can easily be solved using the separation of variables method. This model assumes that the rate of growth of a sample of rice grains by uptake of water is directly proportional to the percentage of rice granules that have not yet absorbed water to capacity. It accounts for the initial height of the rice, the final height of the rice, and a water absorption coefficient.

4.3.1 The Simple Model

$$\frac{dL}{dt} = -h(L - L_c),\ L(0) = 50,\ L_c = 190 \tag{1}$$

In this model, the following things are relevant:

- $\frac{dL}{dt}$ represents the change of volume (height) of rice over time.
- h represents the water absorption rate constant.

- L represents the height of rice at a given time.
- L(0) represents the height at time 0.
- L_c represents the maximum height that the rice will reach during the experiment.

4.3.2 Solving the Model

The model can be solved easily following these steps:

- Separate variables: all terms with L on the left and all terms with t on the right and integrate.
$$\ln(L - L_c) = -ht + C \tag{2}$$
- Exponentiate both sides.
$$L - L_c = Ce^{-ht} \tag{3}$$
- Solve for C.
$$C = \frac{L - L_c}{e^{-ht}} \tag{4}$$
- Use initial conditions from data to solve. The initial condition for this experiment is (0, 50). The height of the rice at time 0 is 50mL.
$$C = \frac{50 - 190}{e^0} = -140 \tag{5}$$
- Solve for L.
$$L = -140e^{-ht} + L_c \tag{6}$$

This is the theoretical model for the results of the experiment.

4.3.3 Analyzing and Applying the Model

Now we must fit the model to our data by finding an h, water absorption coefficient that fits our data.

- Our first step is to find $\frac{dL}{dh}$ by differentiating the last equation.
$$\frac{dL}{dh} = 140te^{-ht} \tag{7}$$
- Now we look at finding the value of h, the water absorption coefficient. Assume that the error between the theoretical model, E, is the difference between the theoretical data and the actual data. We will minimize this error. In this evaluation, we will be using the first three data points: (1,60), (2,62), (3,65).

- The equation we will begin with is:

$$E = \sqrt{\sum_{i=1}^{N}(L_i(l_i) - L_{\text{theory}}(l_i, h))^2}. \tag{8}$$

- Squaring both sides, and minimizing E^2 will yield the same value as minimizing E, we find:

$$E^2 = \sum_{i=1}^{N}(L_i(l_i) - L_{\text{theory}}(l_i, h))^2. \tag{9}$$

- Now in order to minimize the error, E, we differentiate equation (9):

$$\frac{\partial E}{\partial h} = 0 = -\sum_{i=1}^{N} 2(L_i(l_i) - L_{\text{theory}}(l_i, h))\frac{dL_{\text{theory}}}{dh}(l_i, h). \tag{10}$$

- Simplifying, we find:

$$0 = -2\sum_{i=1}^{N}(L_i(l_i) - L_{\text{theory}}(l_i, h))\frac{dL_{\text{theory}}}{dh}(l_i, h). \tag{11}$$

- Using N=3 and the first three data points listed above, plugging in values as follows: $L_i(l_i)$ is the actual data value for N=1, N=2, N=3, $L_{theory}(l_i, h)$ is equation (6), and $\frac{dL_{theory}(l_i,h)}{dh}$ is the derivative found in the first item of this section.

- The resulting equation is as follows:

$$\begin{aligned} 0 = -2[&(60 - (-140e^{-h} + 190))140e^{-h} + \\ &(62 - (-140e^{-2h} + 190))140(2)e^{-2h} + \\ &(65 - (-140e^{-3h} - 190))140(3)e^{-3h}]. \end{aligned} \tag{12}$$

- With much simplification of this equation, we find:

$$0 = -2(140)(-130e^{-h} - 116e^{-2h} - 375e^{-3h} + 280e^{-4h} + 420e^{-6h}). \tag{13}$$

- This equation can be solved for h using Maple, and we find that, with our current data, the water absorption coefficient, h, is approximately equal to .04278min^{-1}.

In order to find h for N equal to the total number of data points, we used a Matlab program that graphed the time values against the log-volume values solved for the ht term, producing a linear relationship. The program also found a linear least squares fit for the new data. To find h, we solved for the slope of the least squares line: $h = .0198\text{min}^{-1}$ The code for this Matlab program can be found in the appendix.

4.3.4 A More Advanced Model

In criticism of our simple model, we can point out that it does not take geometry into consideration. As the rice grain expands with water uptake, there is more surface area available for exposure to, and therefore absorption of, water. Also, our model does not account for a temperature gradient throughout: the center may not be warm enough to gelatinize and absorb water while the outside is. A more complicated model can be borrowed from the work of Landman and McGuinness in their "Modeling the Cooking of a Single Cereal Grain" [4]. It has two parts—a concentration piece and a thermal piece.

4.3.5 The Concentration Piece

To create a more realistic model, we let ρ be the density of water at a particular point inside the rice grain at a given time t:

$$\rho = \rho(\vec{x}, t). \tag{14}$$

Next we determine the change in mass in a control volume V,

$$\frac{d}{dt}\int_V \rho \, dV, \tag{15}$$

the sum of density over a 3D volume, is equal to the exchange at the surface at a given time—the flux. Flux, $\vec{Q}$, will equal mass going into the rice grain through the surface minus the mass going out; mass is conserved. We determine the total flux by taking the dot product of $\vec{Q}$ with the normal vector, then sum over the surface:

$$\frac{d}{dt}\int_V \rho \, dV = \int_S \vec{Q} \cdot n \, dS. \tag{16}$$

Next we apply the Divergence Theorem:

$$\int_S \vec{Q} \cdot n \, dS = \int_V \nabla \cdot \vec{Q} \, dV. \tag{17}$$

So we find:

$$\frac{d}{dt}\int_V \rho \, dV = \int_V \nabla \cdot \vec{Q} \, dV \tag{18}$$

or

$$0 = \int_V (\frac{\partial \rho}{\partial t} - \nabla \cdot \vec{Q}) \, dV. \tag{19}$$

In order for this to be true for any control volume, the integrand must be equal to 0:

$$\frac{\partial \rho}{\partial t} - \nabla \cdot \vec{Q} = 0, \tag{20}$$

or

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \vec{Q}. \tag{21}$$

Now we use a constitutive law—which describes the behavior of a particular material—to eliminate $\vec{Q}$. Our constitutive law states that the flux is proportional to the gradient of the density of water inside a rice grain:

$$\vec{Q} = k\nabla\rho. \tag{22}$$

Substituting, we find:

$$\frac{\partial\rho}{\partial t} = \nabla \cdot k\nabla\rho. \tag{23}$$

4.3.6 The Thermal Piece

Similarly, we can derive an equation for the temperature by applying conservation of energy to a control volume:

$$\frac{\partial T}{\partial t} = \nabla \cdot \vec{Q}. \tag{24}$$

We find:

$$\frac{\partial T}{\partial t} = \alpha\nabla^2 T. \tag{25}$$

Equation (25) is called the heat equation and α is called the thermal diffusivity.

4.3.7 Initial and Boundary Conditions

First, we will consider the concentration piece. At time zero, no water has been absorbed by the rice grain, so our initial condition is:

$$\rho(\vec{x}, 0) = 0. \tag{26}$$

At the boundary of each individual rice grain, there is water covering the surface, so we will say that:

$$\rho(\vec{x}, t) = \rho_0, \tag{27}$$

indicating that there is some amount of water, ρ_0, at the surface of each grain, available for absorption.

The thermal piece works the same way. At time zero, no heat has yet been absorbed by the grain, so its initial temperature is room temperature:

$$T(\vec{x}, 0) = T_0. \tag{28}$$

At the boundary of each grain, the temperature is that of the water surrounding it:

$$T(\vec{x}, t) = T_{\text{BATH}}. \tag{29}$$

Both the concentration and thermal pieces described above, and their initial and boundary conditions, make up this more advanced model of water uptake by rice. Note, in general, the equations for density and temperature are coupled since k from equation (23) depends on temperature.

Note: This more difficult model can be adapted to model the heat transfer within a potato in later sections.

5 Experiment 2: Void Fraction

In this experiment we will be measuring the void space in a particular volume of rice–the empty space between grains–both cooked and uncooked, and how that changes as the volume of rice changes. If the void fraction is different for rice that has been cooked than for uncooked rice, our model from the previous experiment will be affected. Our ultimate goal of the rice experiment is to determine how much water has been taken in by the rice. To determine this, it is important to know the void fraction: if it changes after being cooked, then the way we determine how much water has been taken up must also be changed. We will need all of the equipment we used for the last experiment and additionally:

- 1 100 or 200mL graduated cylinder
- A few extra containers suitable for cooking rice.

5.1 Procedure

Measure 50mL of dry rice and put it into a 200mL beaker. Fill a 100 or 200mL graduated cylinder up to 100mL. Pour water into the dry rice until it reaches the top of the rice, no more. From the water remaining in the cylinder, determine how much water was used to fill the void space in the dry rice. Record "50mL dry" and the amount of water used. Repeat this experiment with the following volumes of rice and record the results: 75mL, 100mL.

Measure 50mL of dry rice and put it into a container that is tall enough to sit in the water bath and marked with, at the very least, approximate volumes. Pour 250mL of water into the container. Place rice in water bath and cook for about one hour. Remove rice from bath. Record the new final volume of the rice. Remove water from the container until the remaining water reaches just to the top of the rice and discard it. Remove the rice from the container, trying to remove as little water as possible. Measure the remaining water. This is the amount of water necessary to fill the void in the cooked rice. Record this data. Repeat this experiment with the same volumes of rice used in the previous part of the experiment.

Void fraction can be determined by dividing the volume of water needed to fill the void of a particular volume of rice by the total volume of the rice. In the case of the cooked rice, use the final volume after cooking in your calculation.

5.2 Results

The first time we performed this experiment, with Vitarroz enriched long grain rice, we found that the void fraction only decreased about ten percent due to cooking (heating and allowing for water absorption). However, when we used Uncle Ben's rice as suggested above (which is supposed to reach four times its size when cooked), we found that the void fraction decreased sixty five percent. This massive drastic drop makes void fraction a serious consideration in a model for Experiment 1. Figure 4 shows how much larger an individual rice grain becomes when it is heated, rather than just soaked.

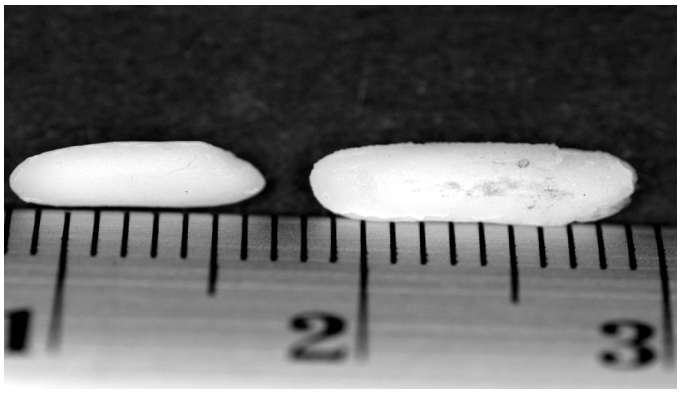

Figure 4: The size of the rice changes drastically, and so does the void fraction.

Figure 5: About half of the starch granules of a potato cooked 40 seconds in the microwave have gelatinized.

6 Experiment 3: Cooking Potatoes

Looking at potatoes is another way to investigate the conditions under which starch gelatinizes. Potatoes are even more interesting than rice because you can see the gelatinized starch versus the crystallized starch visually, and the increase in percentage gelatinized over time. In figure 5 it is easy to see which portions have gelatinized and which have not.

6.1 Part 1: How fast does the inside temperature reach the outside temperature?

In this experiment, we will measure how quickly the potato reaches the temperature of the water bath, both on the surface and at its core. We will need the following materials:

- Water tank and heating utility
- One potato

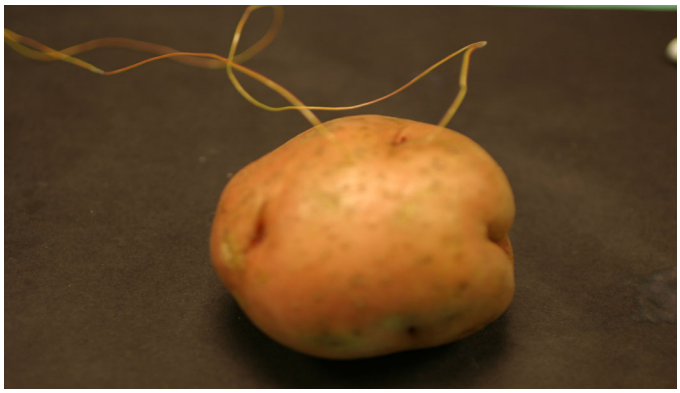

Figure 6: Potato with thermocouples

- Two thermocouples and thermocouple reader
- Stopwatch
- Sharp poker.

This experiment will help us to design any future experiments that investigate cooking potatoes in a water bath at various temperatures.

6.1.1 Procedure

Begin by measuring and recording the mass and surface area of the potato. Heat the water bath to 70°C, as described above. Again, it may take up to one hour. Use your poking device to put a hole into the potato from the top down into the center, making sure not to poke past the center. Slide one thermocouple into the hole, all the way to the center. Poke a tiny hole just under the surface with the stick, and put the tip of the other thermocouple just under the surface. Figure 6 shows a potato with thermocouples inserted. Make sure both thermocouples are fairly secure, and be very careful throughout the experiment not to dislodge them, as you will have to begin the experiment again if the potato must be removed from the bath to reinsert the thermocouples. Carefully place the potato into the preheated bath and start the stopwatch. Record the temperature of the surface of the potato and the inside of the potato at minute intervals until both temperatures reach the temperature of the bath.

6.1.2 Results

Figure 7 displays my results for this experiment. As implied by the model, we have found that heat transfer from the outside to the inside of a potato to be best described as a logistic curve.

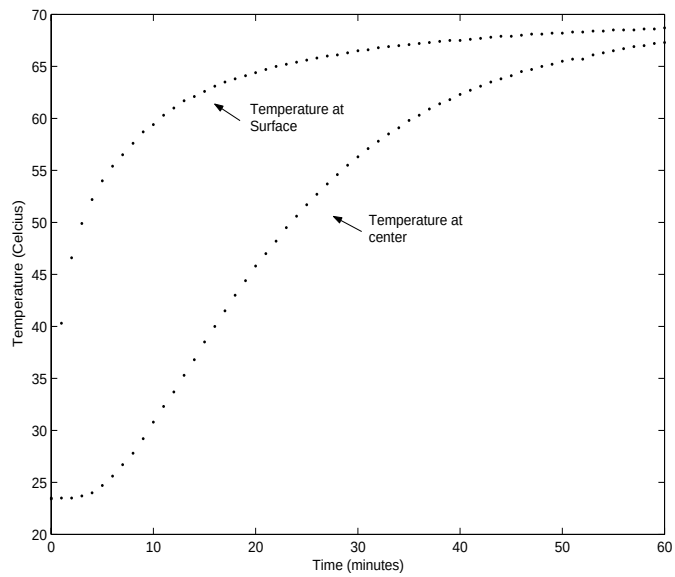

Figure 7: Inside and outside of potato heat up at very different rates, though both can be described as logistic curves.

6.1.3 The Mathematics

Our simple model of heat transfer inside a potato resembles our model for water uptake by rice. This model assumes that the rate of change of temperature is directly proportional to the surface area and to the size of the temperature difference between potato and bath. We begin with:

$$mc_p \frac{dT}{dt} = -hA_s(T - T_B). \tag{30}$$

In this equation, the following things are important to know:

- m is the mass of the potato.
- c_p is the specific heat of a potato.
- $\frac{dT}{dt}$ is the rate of change of temperature over time.
- h is the heat transfer coefficient.
- A_s is the surface area of the potato.
- $(T - T_B)$ is the difference between the temperature at the inside of the potato minus the temperature of its surrounding water bath.

Solving this model as we did our model of rice uptake by water and using our initial condition $T(0) = 23.5$ (inside temperature at time 0), we find:

$$T = -46.7e^{\frac{-hA_s t}{mc_p}} + T_B. \tag{31}$$

To analyze and apply the model,

- Differentiate the last equation with respect to h:

$$\frac{dT}{dh} = \frac{46.7A_s t}{mc_p} e^{\frac{-hA_s t}{mc_p}}. \tag{32}$$

- As before, we now look at finding the value of h, the heat transfer coefficient. In this evaluation, we will be using the first three data points: (1,23.5), (2,23.5), (3,23.7).

- Setup least squares for the new model, and take its derivative to minimize error between our model and the actual data.

$$\frac{\partial E}{\partial h} = 0 = -\sum_{i=1}^{N} 2(L_i(l_i) - L_{\text{theory}}(l_i, h)) \frac{dL_{\text{theory}}}{dh}(l_i, h) \tag{33}$$

- And simplify:

$$0 = -2\sum_{i=1}^{N} (L_i(l_i) - L_{\text{theory}}(l_i, h)) \frac{dL_{\text{theory}}(l_i, h)}{dh}. \tag{34}$$

- Again, using N=3 and the first three data points listed above, and plugging in values as before, the resulting equation is as follows:

$$\begin{aligned} 0 = \frac{-2(46.7)A_s}{mc_p} [&(23.5 + 46.7e^{\frac{-hA_s}{mc_p}} - 70.2)e^{\frac{-hA_s}{mc_p}} \\ &+(23.5 + 46.7e^{\frac{-2hA_s}{mc_p}} - 70.2)2e^{\frac{-2hA_s}{mc_p}} \\ &+(23.7 + 46.7e^{\frac{-3hA_s}{mc_p}} - 70.2)3e^{\frac{-3hA_s}{mc_p}}. \end{aligned} \tag{35}$$

- Substituting values for mass, specific heat, and surface area, and using Maple to solve, we found the heat transfer coefficient, h, to be approximately equal to $.003\frac{J}{min}$.

In order to find h for N equal to the total number of data points, we used a Matlab program that graphed the time values against the log-temperature values solved for the ht term, producing a linear relationship. The program also found a linear least squares fit for the new data. To find h, we solved for the slope of the least squares line: $h = .2239\frac{J}{min}$. This is a reasonable estimate according to the examples we have seen. The Matlab program in the appendix can be modified to solve for h here.

6.2 Part 2: Potatoes and Microwaves

In this experiment, we will study the rate at which potatoes cook in the microwave in terms of percentage cooked over time. We will need the following items:

- Microwave
- Six red potatoes of equal size
- Iodine solution (2 percent)
- Camera
- Knife.

6.2.1 Procedure

This experiment will move quickly. Put one potato into the microwave for 10 seconds. Remove promptly and slice in half. Have a partner focus the camera and be ready to take a picture of the potato after you have applied the iodine. To apply the iodine, unscrew the cap and rub the applicator over the surface of the cross-section, covering the entire surface as quickly as you can. Now have your partner take the picture. You will notice that the iodine turns the uncooked portion of the potato to blue and the cooked portion to orange. Also, the entire potato turns to blue rapidly after applying the iodine, so be sure to take the pictures quickly, while the potato is still orange in uncooked places. Repeat this process, cooking the potatoes at increasing times: 20 seconds, 30 seconds, 40 seconds, 50 seconds, and 60 seconds, taking pictures of each potato after applying iodine to its cross-sectional surface.

Let potatoes sit for an hour or so and you will notice that the portion that was orange before the whole potato turned blue is now cooked-potato-color. Retake the pictures, as the area of cooked potato v. uncooked potato is now better pronounced.

6.2.2 Results

The results of this experiment are somewhat confusing, as originally, the iodine turns the cooked portion of the potato blue, however after sitting for an hour, the outside turns blue and the inside, the cooked portion, turns to a white, cooked-potato color. We are not sure what accounts for the color change.

The idea of this experiment was to measure percentage cooked over time. However, it is very difficult to measure exactly what percent of each potato is cooked. Two-dimensionally, one could use computer software to analyze the picture below, which represents potatoes cooked for 10 seconds, 20 seconds, 30 seconds, 40 seconds, 50 seconds, and 60 seconds from left to right, to determine what percent of the pixels are dark versus light. Also, we cut out images of the potato slices magnified four hundred percent and weighed the white (cooked) portion compared to the entire weight to crudely measure percentage cooked. A graph of these points (time cooked versus percentage cooked) appeared to be roughly linear, although we would assume it to be exponential if a more exact method of measuring percentage cooked were used.

Figure 8 shows potatoes cooked at ten second time intervals after having been stained with iodine solution and sat for one hour.

7 Suggestions for Further Research

- Investigate the role protein plays in gelatinization: if a starch granule contains enough protein, water can be absorbed at room temperature; however, for the amylose and amylopectin to absorb water (the main components of the starch), the granule must be heated. Water absorption causes the proteins to make new bonds and this may affect the gelatinization process [1].

Figure 8: The white portions are cooked while the dark portions are not.

- One suggestion from the food science department was to look at the way additives like sugar or salt affect the speed of water uptake.
- Do experiments on water uptake by rice and heat transfer inside a potato again and see if the original h will fit the new data.
- Perform experiments on water uptake by rice at different temperatures to determine whether or not the water absorption coefficient, h, changes with temperature.
- Additionally, try the water uptake by rice experiment with different types of rice and by cooking in different containers to determine whether h depends on the type of rice or the geometry of its container.
- Incorporate the effects of a change in void fraction on water uptake into the logistic model.
- Try a 2-D version of the void fraction experiment to better understand the way rice packs.
- Try some experiments with single rice grains: cook one rice grain and track its measurements (size increase as in Figure 4) and determine if it follows the data for 50mL of rice grains. Try some rice microscopy.
- Try the heat transfer experiment with potatoes shaped into perfect cubes and spheres.
- Try using a starch indicator rather than iodine in the microwave experiment.

8 Appendix

Code for Matlab program to solve for h in water uptake by rice model:

```
clear;
time=[0:1:60 70:10:170];
```

```
data=[50 60 62 65 68 70 72 77 80 82 85 87 90 90 95 97 98 100 105 108 110 110 110 112 112
112 115 115 117 117 117 120 120 122 122 124 125 126 127 128 128 130 130 132 135 137 137
137 140 140 140 140 140 142 145 145 147 148 148 148 148 155 157 165 170 175 177 177 177
180 185 185];
for i=1:72
logdata(i)=log((190-data(i))/140);
end
plot(time,logdata,'.k')
hold on;
sum1=0;
sum2=0;
for i=1:72
sum1=sum1-time(i)*logdata(i);
sum2=sum2+time(i)*time(i);
end
sum1
sum2
h=sum1/sum2
theory=zeros(72);
for i=1:72
theory(i)=-h*time(i);
end
plot(time,theory,'k');
xlabel('t');
ylabel('Log of data');
```

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